



TECHNOLOGICAL UNIVERSITY (MANDALAY)
DEPARTMENT OF CHEMICAL ENGINEERING
PREPARATION OF SOYBEAN PROTEIN ISOLATE
FROM SOYBEAN MEAL



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ABSTRACT

Soybean protein isolate (SPI) was prepared from defatted soybean meal using alkaline extraction, separation followed by acid precipitation, centrifugation (solid-liquid separation) and drying process.

INTRODUCTION

Soybean meal, a major by-product of soybean oil extraction, contains a high amount of quality plant protein. Soybean protein isolate (SPI) is prepared from defatted soybean meal by separating protein to with excellent functional properties, such as solubility and emulsification. Due to its high nutritional value (protein %) and versatility, SPI is widely used in food processing, health supplements, and various bio-based product applications.

OBJECTIVES

- To prepare the soybean protein isolate (SPI) from defatted soybean meal using alkaline extraction followed by acid precipitation.
- To study the extraction, separation and drying processes.

Process Flow Chart

Raw Material

Soaking

Alkaline Extraction
(at 70C°- 80°C for 45 minutes)

Solid-Liquid Separation
(for 10 minutes)

Acid Precipitation
(for 15 minutes)

Solid-Liquid Separation
(for 10 minutes)

Washing

Drying
(at 55°C for 18 hours)

Soybean Protein Isolate (SPI)

Experimental Work



Soybean Meal



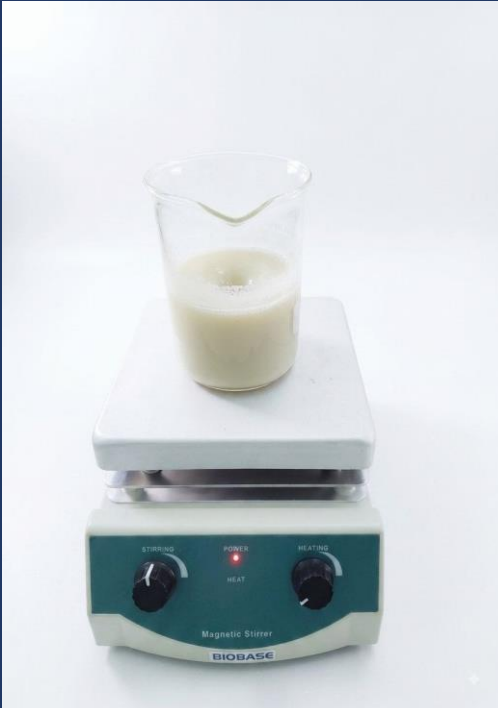
Soaking



Alkaline Extraction



Separation



Acid Precipitation



Solid-Liquid Separation



Washing



Drying



Soybean Protein Isolate (SPI)

CONCLUSION

- Soybean protein isolate (SPI) was prepared from defatted soybean meal using alkaline extraction, separation, acid precipitation, solid-liquid separation, washing and drying at 55°C for 18 hours.
- Utilizing soybean meal not only provides a cost-effective raw material for health supplements but also promotes sustainable waste management in industrial processing.